LoRa-Enabled Smart Cucumber Polytunnel System for Disease Detection, Fertigation and Water Stress Management in Sri Lanka

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***Abstract*—This paper presents a LoRa-enabled smart polytunnel system designed to support automated monitoring and management of cucumber cultivation. The proposed system integrates a Radio Frequency Identification (RFID) based autonomous robot, deep learning-based disease and stress detection, a nutrition management module, and a long-range communication network within a unified framework. The robotic platform navigates through crop rows, captures plant images, and performs targeted fungicide spraying based on real-time disease detection. A hybrid computer vision approach combining YOLOv8 and Mask Region-based Convolutional Neural Network (Mask R-CNN) is used to identify powdery mildew, downy mildew, and water stress conditions, enabling timely intervention and reducing unnecessary chemical usage. In addition, a nutrition management module incorporating real-time Nitrogen, Phosphorus, and Potassium (NPK) sensing and automated mixing supports adaptive fertigation, while flower detection is used as biological trigger for stage-based nutrient control. The system also includes cucumber detection and ripeness estimation using image-based measurements combined with ultrasonic sensing. A LoRa-based communication layer ensures reliable data transmission and supports environmental monitoring with sensor anomaly detection. Experimental evaluation conducted in a real polytunnel environment demonstrated high detection accuracy, reliable navigation, and approximately 80% reduction in fungicide usage. The results highlight the potential of the proposed system as a low-cost and scalable solution for precision agriculture in resource-constrained environments.**

Keywords—smart agriculture, polytunnel automation, cucumber disease detection, cucumber cultivation

I. INTRODUCTION

Agriculture remains an important sector in Sri Lanka, contributing to food security, rural livelihoods, and economic stability. Among vegetable crops, cucumber *(Cucumis sativus L.)* is widely cultivated due to its high demand and commercial value. In recent years, protected cultivation methods such as polytunnels have become increasingly important for improving crop productivity and managing environmental limitations in Sri Lanka [1], [2].

However, cucumber cultivation in polytunnels still faces major challenges, especially from diseases such as downy mildew and powdery mildew as well as water stress conditions that can negatively affect plant growth, yield and quality [3], [4]. In most cases, these problems are identified through manual observation, which may delay treatment and reduce the effectiveness of management practices. As a result, farmers may experience higher production costs, unnecessary chemical use and reduced productivity.

To address these challenges, smart agriculture technologies provide a more efficient approach through the integration of machine vision, automation, sensing and communication systems. In this context, this research proposes a LoRa-enabled smart polytunnel solution for Sri Lankan cucumber farmers. The system integrates robotic monitoring, image-based disease and stress detection, precise fertigation support and long range communication to support early detection and timely intervention.

The proposed solution is expected to improve crop health monitoring, reduce resource waste, minimize unnecessary fungicide application and support more sustainable cucumber cultivation in Sri Lanka.

II. LITERATURE REVIEW

Protected cultivation has gained attention in Sri Lanka as an effective way to improve vegetable production under controlled conditions. In crops such as cucumber, polytunnel systems help maintain favourable microclimates and improve yield performance. Previous local studies shown that integrating IoT-based monitoring and control into protected-house cultivation can significantly enhance productivity by managing environmental factors such as temperature, humidity and irrigation more efficiently [1], [2].

With the advancement of smart agriculture, IoT-based systems have been widely used to automate greenhouse operations. These systems typically rely on sensors and communication networks to collect environmental data and support real-time decision making [3]. However, most existing implementations mainly focus on environmental control and fertigation, with limited attention given to direct plant-level monitoring and early detection of diseases.

In recent years, machine vision and deep learning techniques have shown strong potential for plant disease detection. Studies have demonstrated that convolutional

neural networks can accurately identify diseases such as downy mildew and powdery mildew from leaf images, even under complex field conditions [4], [5]. Early detection of these diseases is important, as they can spread rapidly and significantly reduce crop yield and quality.

In addition to diseases, water stress is a key factor affecting cucumber growth. However, compared to disease detection, fewer studies have focused on identifying water stress using image-based methods. Existing research highlights that visual symptoms of stress and disease can often overlap, making more accurate detection more challenging [6]. Recent reviews have also highlighted the potential of AI techniques for crop water stress detection and plant stress monitoring [7], [8]. This indicates the need for systems that can identify both biotic and abiotic stress conditions within a single framework.

Furthermore, reliable communication is essential for smart agriculture systems. Technologies such as LoRa have been widely adopted due to their low power consumption and long-range communication capabilities, making them suitable for polytunnel environments with multiple sensors and devices [3].

Overall, while previous studies have explored protected cultivation, IoT-based automation, and image-based disease detection separately, there is limited work on integrating these components into a single system. In particular, there is a gap in solutions that combine plant level image analysis, disease and water stress detection and real-time communication within cucumber polytunnels. This research aims to address that gap by proposing an integrated smart polytunnel system tailored to Sri Lankan conditions.

III. METHODOLOGY

The proposed system consists of multiple integrated modules designed to enable automated monitoring and management of cucumber plants in a polytunnel environment. These modules include an Radio Frequency Identification (RFID) based autonomous robot, a disease and water stress detection module, a nutrition management module, a cucumber flower detection module, a cucumber detection and ripeness estimation module and a LoRa-based communication system with sensor anomaly detection. The overall system operates through coordinated sensing, image analysis, communication and automated actuation. Fig. 1. shows the overall system architecture of the proposed system.

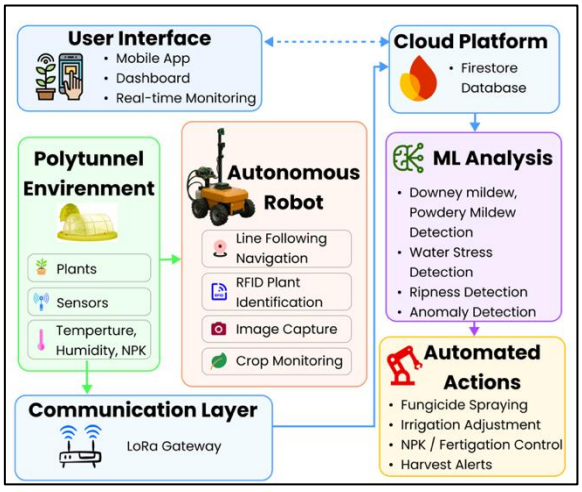


Fig. 1. System diagram of the proposed system

A. RFID-Based Autonomous Precision Spraying Robot

a) Mechanical Architecture and Kinematics



Fig. 2. Developed RFID-based autonomous precision spraying robot

The autonomous robotic platform was designed to operate within the limited space of Sri Lankan cucumber polytunnels. The chassis dimensions are 30cm in width, 35cm in length and 70cm in height. Locomotion is provided by a four-wheel-drive system using four DC gear motors coupled with 8cm wheels. Two BTS7960 motor drivers are used to control the motors, providing sufficient torque to move across uneven agricultural terrain at a maximum motor speed of 105rpm.

b) Autonomous Navigation and Spatial Localization

For row navigation, the robot uses a continuous line following mechanism. Two analog infrared sensors mounted on the underside detect a high contrast track, allowing the ESP32 microcontroller to adjust the differential motor speeds dynamically. This closed loop control maintains a lateral tracking deviation of approximately $\pm$1cm. For plant level localization, the robot uses a RFID system. Passive 13.56MHz RFID tags are placed at the base of each cucumber plant, and an RC522 RFID reader with a 3cm reading range is mounted on the robot. When a tag is detected, the corresponding plant identifier is read, and a hardware interrupt stops the drive motors to begin the scanning process.

c) Motorized Vertical Scanning Array

Since cucumber vines grow vertically, a motorized scanning mechanism was developed to capture the full canopy profile. The mechanism consists of a 0.5m threaded rod and a 0.75m smooth guide rod driven by a NEMA 12 stepper motor. An A4988 stepper driver configured for 1/16 micro stepping provides high-resolution vertical movement of the camera carriage. During each scanning cycle, which takes about 5 to 10 minutes per plant, the carriage pauses at predefined intervals to capture 15 high resolution images (1280x720) using a mobile optical sensor.

d) Deep Learning Architecture

The vision-based analysis module uses a hybrid deep learning approach for disease detection, water stress identification, and severity estimation. First, YOLOv8 is used to detect and localize cucumber leaves from images captured by the robot-mounted camera. The detected leaf regions are then passed to a Mask R-CNN-based model to classify the leaf condition as

healthy, powdery mildew, downy mildew, or water stress. The model also supports severity estimation by analyzing the affected area of the leaf. The models were trained using augmented cucumber leaf images collected under polytunnel conditions, and performance was evaluated using precision, recall, and F1-score. Based on the final prediction, the system activates the relevant fungicide pump, sends a water stress notification, or takes no action for healthy leaves.

e) Precision Spraying Actuation and Power Distribution

Targeted fungicide spraying is carried out using a 5 V DC liquid pump controlled through an L298N motor driver. When the system detects a diseased leaf, the pump is activated for 10 seconds to spray only the infected foliar region instead of the entire plant. If downy mildew is detected, Pump 1 is activated. If powdery mildew is detected, Pump 2 is activated. If water stress is detected, no spraying is performed and only a notification is sent. The robotic platform is powered by four 3.7V 18650 lithium-ion batteries connected in series, providing a nominal 12V supply with a capacity of 5000mAh. Step down buck converters generate isolated 5V outputs for the control logic, sensors and pump enabling a continuous operating time of approximately 2 hours.

B. Disease Detection and Water Stress Detection

The disease detection and water stress detection module were developed to analyze cucumber leaf images captured by the robot mounted camera and identify plant health conditions in real time. The target classes considered in this study include powdery mildew, downy mildew, water stress and healthy leaves. In addition to basic classification, the module also estimates disease severity to determine how much of the leaf is affected, enabling more precise and targeted responses. When a disease is detected, the system activates the fungicide sprayer mounted on the robot, while water stress detection results in a notification being sent to the farmer through the monitoring platform.

To achieve this, a hybrid deep learning approach is used. Initially, a YOLOv8 object detection model is applied to detect and localize cucumber leaves from the captured plant images. This step helps isolate the leaf region from surrounding elements such as stems, flowers and background noise allowing the model to focus only on relevant features for disease analysis. Each detected leaf region is then extracted and passed to the classification stage. The extracted leaf image is analyzed using a Mask Region-based Convolutional Neural Network (Mask R-CNN) based classification model, which determines the health condition of each leaf

Beyond classification, the system evaluates disease severity by estimating the proportion of infected areas within each detected leaf. This allows the system to distinguish between early stage and more advanced infections, supporting better decision making in practical field conditions. Based on the predicted class, the system determines the appropriate response. If the output corresponds to powdery mildew or downy mildew, the robot activates the fungicide spraying mechanism. The spraying mechanism is designed to apply fungicide only to the infected leaf regions for a specific duration, reducing unnecessary chemical usage and protecting healthy parts of the plant. In contrast, when water stress is detected, the system sends a notification to the farmer through the platform. If the plant is identified as healthy, no action is taken. This approach ensures that detection results are not only informative but also actionable, enabling timely intervention in the field while avoiding unnecessary treatments. The workflow of the detection module is illustrated in Fig. 3.

The performance of the model is evaluated using standard classification metrics. Precision, recall and F1-score. These metrics are used to evaluate how effectively the model detects each condition and how reliable it is for real-time decision

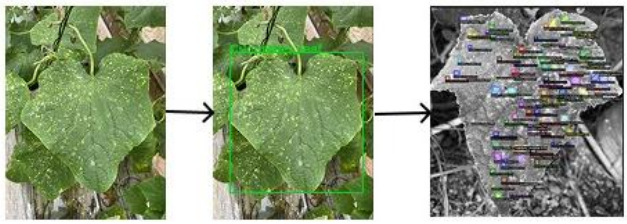


Fig. 1. Workflow of the disease and water stress detection module

making within the polytunnel system.

C. Nutrition Management Module

The nutrition management module monitors soil nutrient conditions and automates fertigation based on crop growth stages. It integrates two IoT subsystems, namely the Nitrogen, Phosphorus, and Potassium (NPK) Monitor and the NPK Mixer, along with a flower detection model deployed on the robotic platform. The main objective is to maintain optimal nutrient availability while supporting both automated and manual control through the mobile application.

The NPK Monitor measures real-time soil macronutrient levels (N, P, K) near the root zone and transmits the data to the central system for remote monitoring. These measurements also serve as feedback to evaluate nutrient levels before and after fertigation. The NPK mixer prepares nutrient solutions according to predefined ratios corresponding to crop growth stage. Nutrient dispensing is controlled using solenoid pumps, while load cell sensors measure the delivered quantities to ensure dosing accuracy. The prepared solution is then supplied through the drip irrigation network for uniform distribution.

The nutrient management strategy follows a hybrid approach combining rule-based scheduling and event-driven control. Early growth stages operate on predefined schedules while the flowering stage is triggered through image-based flower detection. Based on the active stage, nutrient ratios are adjusted dynamically. A feedback mechanism compares target and measured nutrient values to validate fertigation performance and improve dosing precision. The system continuously updates nutrient conditions and allows real-time monitoring and manual intervention through the mobile application.

D. Cucumber Flower Detection

To identify the flowering stage of cucumber plants inside the polytunnel, an image-based detection model was developed using YOLOv8 framework. The main purpose of this model is to detect cucumber flowers from images captured by the robot and use this information as a biological trigger to switch the nutrition management system to the flowering stage. This allows the system to move from fixed scheduling to plant condition based nutrient control. The dataset used for training consisted of 798 images collected under polytunnel

conditions. These images were annotated using the Roboflow platform by marking visible flowers with bounding boxes. Data preprocessing and augmentation techniques, including resizing, normalization, rotation, flipping, brightness adjustment and zooming were applied to improve model robustness under varying real-world conditions.

The flower detection model was trained using YOLOv8, which detects objects in a single pass and gives each detection a confidence score. Only detections above a selected confidence threshold were accepted as flowers. The model was trained by minimizing the YOLO loss function, and its performance was evaluated using precision, recall, F1-score, and mAP@0.5. In addition, mean average precision (mAP@0.5) is used as a key performance indicator to evaluate detection quality. After training, the model is integrated into the robotic vision system. As the robot scans plants, captured images are processed to detect flowers. When at least one flower is detected, the system identifies this as the onset of the flowering stage and updates the nutrition management process accordingly.

This integration enables continuous and non-destructive monitoring of crop development while improving the adaptability of nutrient scheduling. By linking flower detection with fertigation control, the system responds directly to plant conditions rather than relying solely on predefined timelines.

E. Cucumber Detection and Ripeness Estimation

The cucumber detection and ripeness estimation module were developed to identify cucumbers from plant images and determine harvest readiness based on estimated physical size. During plant scanning, a camera mounted on the robot captures cucumber images, while an ultrasonic sensor measures the distance between the camera and the plant. This distance information is combined with image-based measurements to estimate the actual cucumber size.

A YOLOv8 object detection model trained on 891 cucumber images is used to detect cucumbers in the captured images. For each detected cucumber, the model returns a bounding box from which the length and width in pixel units are obtained. In this approach, the ultrasonic sensor provides depth information, while the image provides pixel-based measurements for conversion into real-world dimensions. The ripeness decision is based on standard market size requirements for salad cucumber in Sri Lanka. A cucumber is considered ripe when its estimated length is between 15-16cm, and its diameter is between 2.5-3.0cm. This condition is defined as

$$R = \begin{cases} 1, & 15 \le L_{cm} \le 16 \; and \; 2.5 \le W_{cm} \le 3.0 \\ 0, & otherwise \end{cases} \quad (1)$$

Where $R = 1$ indicates a ripe cucumber and $R = 0$ indicates a non-ripe cucumber. L_{cm} and W_{cm} represent the estimated real-world cucumber length and diameter in centimeters. This method provides a practical and reliable way to estimate cucumber maturity by combining distance sensing with image-based object detection.

F. LoRa Communication and Sensor Anomaly Detection

The LoRa communication and sensor anomaly detection module was developed to provide reliable long-range environmental monitoring within the polytunnel. Temperature and humidity sensors are deployed as distributed sensing nodes, and their readings are transmitted via LoRa to a gateway and cloud database. LoRa is selected due to its long communication range, low power consumption and suitability for agricultural environments where continuous Wi-Fi connectivity is limited.

Once sensor data are received, the system performs anomaly detection to identify abnormal environmental conditions or potential sensor faults. Let $T_1 \; and \; T_2$ denote temperature readings from two sensor nodes and $H_1 \; and \; H_2$ denote the corresponding humidity values. The disagreement between sensors is computed as

$$\Delta T = |T_1 - T_2| \quad (2)$$

$$\Delta H = |H_1 - H_2| \quad (3)$$

Where $\Delta T \; and \; \Delta H$ represent the absolute differences in temperature and humidity readings, respectively.

These values are then compared against predefined thresholds to determine the presence of anomalies. If the differences exceed predefined thresholds, an anomaly condition is triggered. This approach enables detection of abnormal deviations caused by sudden environmental changes or inconsistent sensor behavior. The sensor readings and anomaly status are transmitted to the cloud database and made accessible through the mobile application for real-time monitoring and alerts. This design improves the reliability of environmental sensing and provides an additional decision support layer for effective polytunnel management.

IV. RESULTS AND DISCUSSION

A. Experimental Setup

The proposed system was evaluated in a commercial cucumber polytunnel located in Sri Lanka. The experimental setup consisted of five cultivation rows, each measuring 8m in length, with a row spacing of 150cm and an inter-plant spacing of 70cm. A total of 20 plants were selected randomly for testing. All experiments were conducted under real operating conditions to assess the performance of the integrated system.

B. RFID-Based Autonomous Precision Spraying Robot Performance

1) Kinematic and Localization Performance

During Field operation, the robot demonstrated stable navigation despite the high humidity conditions inside the polytunnel. The differential drive control maintained lateral deviation within approximately ±1cm, allowing the robot to follow rows accurately.

The RFID-based localization mechanism also performed reliability. Although the RC522 module typically provides a theoretical detection accuracy between 70% and 90% at operating speeds, the system achieved a 100% plant identification rate during testing. In practical, the robot was able to stop at every designated plant without missing any scanning point. This suggested that the integration of RFID with controlled navigation significantly improves real-world reliability.

2) Actuation and Resource Optimization

The precision spraying mechanism proved effective in reducing unnecessary chemical usage. Instead of applying fungicide uniformly across the plant, the system activated the pump only when disease was detected. As result, fungicide consumption was reduced by approximately 80% compared to conventional spraying methods. This highlights the practical benefit of combining AI-based detection with targeted actuation. The field results are summarized in Table I.

TABLE I : FIELD PERFORMANCE OF ROBOTIC SYSTEM COMPONENTS

System Component	Field Result
Navigation Stop Accuracy	100%
Line Tracking Deviation	±1 cm
Spray Target Accuracy	80%
Agrochemical Reduction	80%
Continuous Runtime	2 hours

C. Disease Detection and Water Stress Detection Performance

The disease detection module showed consistently strong performance across all evaluated classes. Both powdery mildew and downy mildew achieved F1-score above 0.95, indicating a good balance between precision and recall. The results for water stress were particularly notable, with a recall and F1-score of 1.0000. this indicates that water stressed samples in the test set were successfully identified. This enables early intervention for water stress management. Table II shows the performance metrics of the disease and water stress detection model.

TABLE II : PERFORMANCE METRICS OF DISEASE AND WATER STRESS DETECTION MODEL

Class	Precision	Recall	F1-Score	Accuracy
Powdery Mildew	0.9000	0.9091	0.9045	0.9091
Downy Mildew	0.9000	0.9130	0.9065	0.9130
Water Stress	0.9000	1.0000	0.9474	1.0000

An important strength of this module is its direct connection to field actions. When a disease is detected, the system triggers targeted fungicide spraying. In contrast, water stress detection results in a notification to the farmer rather than automatic spraying. However, the performance may still be influenced by factors such as lighting variations, overlapping leaves, and similarities between early disease symptoms and the stress conditions. In addition, the perfect results observed for water stress should be interpreted with caution, as they may be affected by dataset size and variability. Further validation with larger and more diverse datasets would improve confidence in the model's generation.

D. Nutrition Management Module

Nutrient management module successfully demonstrated real-time monitoring, automated nutrient mixing and controlled delivery through the drip irrigation system. The integration with flower detection allowed the system to automatically switch to the flowering-stage nutrient profile, improving adaptability.

In addition, the mobile application enabled farmers to monitor nutrient conditions and manually adjust system parameters when needed. This combination of automation and user control makes the system more practical for real-world deployment. However, the current results mainly confirm functional feasibility, The overall performance is still influenced by factors such as sensor accuracy, mixing precision and irrigation uniformity. Further work is required to calibrate the system and validate nutrient concentrations under different field conditions.

E. Cucumber Flower Detection

The flower detection model achieved satisfactory performance under real polytunnel conditions. Detecting flowers can be challenging due to their small size and frequent occlusion by leaves, yet the model was able to identify flowering stages with reasonable accuracy. Table III presents the evaluation results of the flower detection model.

TABLE III : PERFORMANCE METRICS OF CUCUMBER FLOWER DETECTION MODEL

Metric	Value
Precision	0.925
Recall	0.886
mAP@0.5	0.941
mAP@0.5:0.95	0.862
F1-Score	0.905

This module is particularly important because it serves as a biological trigger for nutrient management. Instead of relying solely on time-based scheduling, the system adapts nutrient delivery based on actual plant development. This improves the overall responsiveness and efficiency of the fertigation process. Nevertheless, performance may still be affected by environmental conditions such as lighting variations, motion blur and uneven flowering patterns. Future improvements could focus on expanding the dataset and enhancing detection stability under varying field conditions.

F. Cucumber Detection and Ripeness Estimation

The cucumber detection model demonstrated strong performance under polytunnel conditions. Table IV shows the performance metrics of cucumber detection and ripeness estimation model.

TABLE IV : PERFORMANCE METRICS OF CUCUMBER DETECTION AND RIPENESS ESTIMATION MODEL

Metric	Value
Precision	0.9190
Recall	0.9024
mAP@0.5	0.9697
mAP@0.5:0.95	0.8704

The high mAP@0.5 value (0.9697) indicates that the model was able to accurately localize cucumber objects within the captured images. Beyond detection, this module plays a key role in estimating fruit maturity. By combining image-based measurements with distance data from the ultrasonic sensor, the system estimates the actual size of each cucumber. Based on these measurements, cucumbers are classified as ripe if they fall within the predefined size range.

This combined approach provides a more practical solution compared to relying solely on visual appearance. However, the accuracy of ripeness estimation depends not only on detection performance but also on factors such as

sensor precision, camera calibration and viewing angle. Small variations in these parameters can affect the estimated dimensions. Despite these limitations, the module provides a

reliable foundation for automated monitoring or harvest readiness.

G. LoRa Communication and Sensor Anomaly Detection

The LoRa-based communication system enabled reliable transmission of environmental data from distributed sensor nodes to the cloud platform. This is particularly important in agricultural settings where continuous Wi-Fi coverage may not be available.

The anomaly detection mechanism, based on comparing readings from multiple sensors, helped identify potential inconsistencies in temperature and humidity data. This improves system reliability by reducing the likelihood of incorrect decisions caused by faulty sensor readings.

One of the key advantages of this approach is its simplicity and low computational cost, making it suitable for real-time deployment. However, its effectiveness depends on the selection of appropriate threshold values. If thresholds are too strict, false alarms may occur, while overly relaxed thresholds may fail to detect meaningful anomalies. Therefore, further tuning and long-term evaluation are necessary.

H. Overall Discussion and Limitations

Overall, the results demonstrate that the proposed system successfully integrates robotics, computer vision and IoT technologies into a practical smart agriculture solution. The system performs well in real polytunnel conditions, showing strong capabilities in detection accuracy, navigation reliability and resource optimization.

At the same time, several limitations were identified. The current scanning process, which takes approximately 5-10 minutes per plant, limits scalability for larger farms. In addition, reliance on external computing for AI inference introduces latency and reduces system independence.

Future improvements should focus on increasing scanning speed, improving hardware efficiency and deploying edge-based AI models directly on the robot. Despite these challenges, the system provides a strong foundation for automated, data-driven cucumber cultivation in polytunnel environments.

V. CONCLUSION

This study presented a smart polytunnel system for cucumber cultivation by integrating robotics, computer vision and IoT technologies. The system combines an RFID-based autonomous robot, disease and water stress detection, nutrient management and LoRa-based communication to enable automated monitoring and decision-making.

The results show that the system performs effectively under real polytunnel conditions. The detection models achieved high accuracy, supporting reliable identification of diseases, water stress and crop development stages. In addition, the precision spraying mechanism significantly reduced agrochemical usage, while the ripeness estimation and flower detection modules improved crop monitoring and nutrient management. However, some limitations remain, including the time required for plant scanning and the dependence on external computing for model inference. Future work will focus on improving system speed, enhancing robustness and enabling edge-based processing for fully autonomous operation. Overall, the proposed system demonstrates a practical and scalable approach to precision agriculture, particularly for resource-constrained farming environments.

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